

UNIT 11

INDUSTRIAL CHEMISTRY



Q #01. Give the scope of pharmaceutical industries in Pakistan?**SCOPE OF PHARMACEUTICAL INDUSTRIES IN PAKISTAN:****1) Overview of the Pharmaceutical Industry:**

- The pharmaceutical industry in Pakistan plays a crucial role in the healthcare sector.
- It involves the development, manufacturing, and distribution of drugs and medicines.

2) Economic Contribution:

- One significant aspect of the industry is its economic contribution to the country.
- It creates job opportunities and contributes to the overall GDP.

3) Employment Opportunities:

- The pharmaceutical sector provides a wide range of job opportunities for individuals with diverse skill sets.
- Careers include research and development, manufacturing, quality control, and marketing.

4) Research and Development (R&D):

- Pharmaceutical companies invest in research and development to discover new drugs and improve existing ones.
- This fosters innovation and contributes to global healthcare advancements.

5) Access to Essential Medicines:

- The industry ensures a consistent supply of essential medicines to the public.
- This contributes to the overall well-being of the population.

Q#02. What is antihistamine drug? Mention the symptoms in which they use?

Antihistamine Drugs: Antihistamines are a type of medication used to treat various symptoms caused by the release of histamine in the body.

What is Histamine?

- Histamine is a natural substance released by the body during an allergic reaction.
- It causes symptoms like itching, sneezing, and runny nose.

How Antihistamines Work:

- Antihistamines block the effects of histamine, reducing or alleviating allergy symptoms.

Common Symptoms Treated:

- Antihistamines are often used to address a range of allergy-related symptoms.

Q #03. Write the names of two synthetic and two natural polymers.**1) Synthetic Polymers:**

Synthetic polymers are human-made and widely used in everyday products.

(i) Polyethylene:

Description: Polyethylene is a common synthetic polymer with various applications.

Common Uses: It is used in plastic bags, containers, and even some types of clothing.

(ii) Polyvinyl Chloride (PVC):

Description: PVC is another synthetic polymer known for its versatility.

Common Uses: PVC is used in pipes, cables, and construction materials.

2) Natural Polymers:

Natural polymers are derived from living organisms and have diverse applications.



(i) Starch:

Description: Starch is a natural polymer found in plants and serves as an energy storage molecule.

Common Uses: It is commonly used in the food industry as a thickening agent.

(ii) Proteins:

Description: Proteins, like silk and wool, are natural polymers with complex structures.

Common Uses: Silk is used in textiles, while wool is used in clothing and textiles.

Q # 04. Write the name of four main components of Nail polish?**COMPONENTS OF NAIL POLISH:**

1) Solvents: Solvents are liquids that dissolve other substances to create a smooth and applicable nail polish texture.

2) Film Formers: Film formers create a sturdy layer on the nail as the polish dries, providing the desired color and finish.

3) Pigments and Colorants: Pigments and colorants provide the vibrant hues and shades to the nail polish.

4) Plasticizers: Plasticizers maintain the flexibility of the nail polish, preventing it from becoming brittle.

Q #05. How can you define Cosmetics? Describe four cosmetics which are commonly used.**COSMETICS:**

Definition of Cosmetics: Cosmetics are products designed for enhancing or altering a person's appearance. They include a wide range of items used for skincare, makeup, fragrance, and hair care.

1) Lipstick: Lipstick is a popular cosmetic applied to the lips for color and texture.

Ingredients:

- **Wax and Oils:** Provide texture and consistency.

- **Pigments:** Add color to the lips.

Common Uses:

- Enhancing lip color.

- Creating different lip shapes and styles.

2) Nail Polish: Nail polish is a cosmetic product applied to the nails for color and protection.

Components:

- **Solvents:** Assists in drying and application.

- **Film Formers:** Create a durable layer on the nails.

Common Uses:

- Adding color and style to nails.

- Protecting nails from damage.

3) Hair Dyes: Hair dyes are cosmetics used to change or enhance hair color.

Ingredients:

- **Ammonia or Peroxide:** Lighten or lift existing hair color.

- **Coloring Agents:** Add the desired color to the hair.

Common Uses:

- Changing hair color for fashion.

- Covering gray hair.

4) Perfumes: Perfumes are fragrant liquids applied to the body for a pleasant scent.

Components:

- **Essential Oils:** Provide the fragrance.

- **Alcohol or Water:** Dissolve and disperse the fragrance.

Common Uses:

- Enhancing personal scent.
- Creating a pleasant atmosphere.

Q #06. What are pesticides? Explain various types of pesticides along with their specific use.**PESTICIDES:**

Definition of Pesticides: Pesticides are substances or chemicals used to control, repel, or eliminate pests that can harm crops, livestock, or humans.

Types of pesticides:

1) Insecticides: Insecticides target and control insect pests that can damage crops.

Types:

- **Organophosphates:** Effective against a wide range of insects.
- **Pyrethroids:** These are used for mosquitoes and other pests.

Specific Use:

- Protecting crops from insect damage.
- Controlling disease-carrying insects.

2) Herbicides: Herbicides are designed to control or eliminate unwanted plants, commonly known as weeds.

Types:

- **Selective Herbicides:** Target specific types of plants.
- **Non-Selective Herbicides:** Kill a wide range of plants.

Specific Use:

- Weed control in agriculture.
- Clearing unwanted vegetation in gardens.

3) Fungicides: Fungicides combat fungal diseases that can harm crops or plants.

Types:

- **Protective Fungicides:** Applied before the onset of disease.
- **Curative Fungicides:** Used to treat existing fungal infections.

Specific Use:

- Preventing and treating fungal diseases in crops.
- Preserving the quality of fruits and vegetables.

